

United States Gym Data Report

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To Whom It May Concern:

Thank you for trusting our team to use the CRISP-DM method in analyzing your gym membership dataset. Our team conducted a detailed analysis of your company based on membership data over the past year. Using the dataset, we were able to understand churn behavior, identify meaningful predictors, and develop machine learning models to support retention strategies in the future.

Business Understanding

The primary business goal in this analysis is to reduce customer churn by understanding the key drivers behind the cancellation of gym memberships. By applying Python learning techniques, our objective is to predict which members are most at risk of churn and provide actionable insights for retention efforts in the future.

Data Understanding

The dataset contains 4000 rows and 14 columns. The key variables that our team is using to complete this analysis include tenure, monthly charges, total charges, usage frequency, and demographic information, which are used to predict whether the customer churned (cancelled their membership). These variables allow us to study behavioral and financial patterns associated with what makes a customer cancel their gym membership.

Data Preparation

We had to first clean the dataset to be able to complete an accurate analysis to help your company take a step forward in membership retention. Using the `.info()` function, we determined that the dataset had no null values that needed to be filled. Next, it was important to examine how the churn variable is distributed. Table 1 shows us that 2,939 people renewed their membership and 1061 canceled their membership. This means that nearly 75 percent of customers chose to stay with the gym, giving us a clear baseline for retention before any modeling or deeper analysis.

Next, we removed `Avg_class_frequency_current_month` after realizing it introduced data leakage. Since it reflects behavior from the same month in which churn occurs, it artificially inflated the relationship with churn. When we checked correlations using `.corr()`, this feature showed an unusually strong link to churn, which confirmed that it was leaking outcome information and distorting the analysis.

After the initial review, we converted several numerical columns into binned categorical groups to make the patterns easier to interpret and more useful for modeling. Lifetime, Age, and

Month_to_end_contract were all transformed into bins because their raw numerical values did not provide meaningful separation, while grouped ranges capture clearer behavioral differences. For lifetime, we created “0_6_months,” “7_12_months,” “13_24_months,” and “25_plus_months.” Age was grouped into “18_24,” “25_29,” “30_34,” and “35_plus,” allowing your gym to evaluate churn trends by demographic. Month_to_end_contract was converted into contract_end_group to highlight how churn changes as members approach the end of their contract. This also prevents the model from overfitting to exact month values that are not predictive. We then reviewed the distribution of each grouped feature, shown in Table 2. Most cancellations occurred in the 1 to 3 month window, with 2,244 members in that range. All categorical columns were then converted using get_dummies so the variables could be used by machine learning models, while dropping the original numerical columns.

Figure 1: Count of Gym Members by Churn Status (0 = Retained, 1 = Churned)

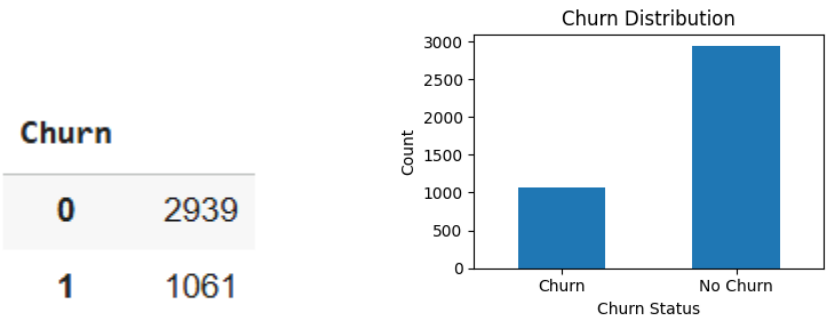
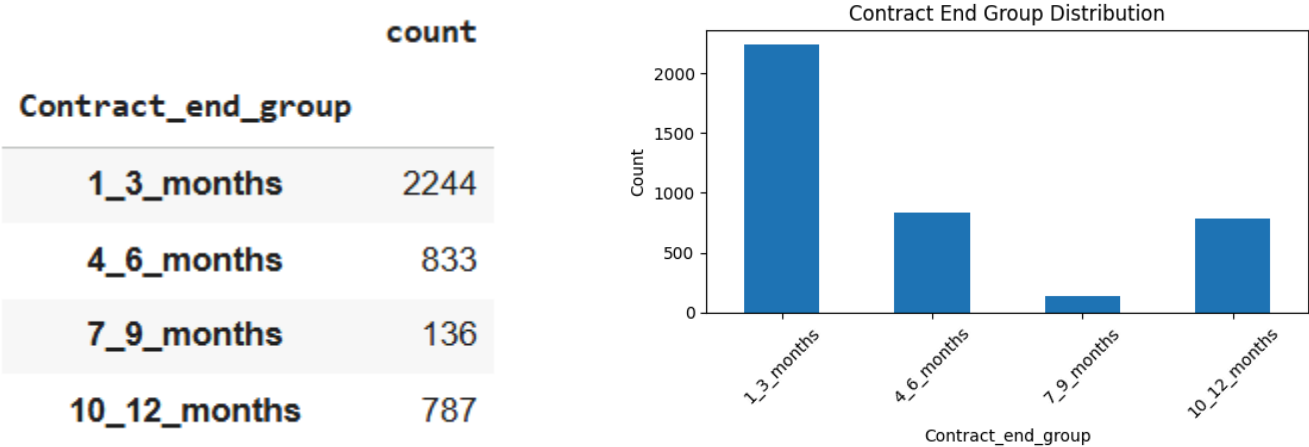


Figure 2: Count of Gym Members by Remaining Contract Duration



Modeling

The first step of modeling was to identify the Churn variable and predictors. The predictor variables were Income, Days_since_last_visit, Avg_class_frequency_total, Avg_PT_total, Total_charges, Lifetime_group, Age_group, Contract_end_group, and Contract_period. The target variable was Churn (1 = churned, 0 = retained). We then split the dataset into training and testing sets, 70/30. We used 4 different machine learning models, which were a Logistic Regression, a Random Forest Classifier, a Decision Tree, and an XGB Classifier. Using multiple algorithms allowed us to compare predictive performance and identify the best model for predicting churn. Each model learned patterns linked to churn. Logistic Regression estimated feature effects, while the tree-based models (Decision Tree, Random Forest, and XGBoost) learned predictive splits in different ways. Accuracy was measured with confusion matrices; we did not use any unsupervised methods.

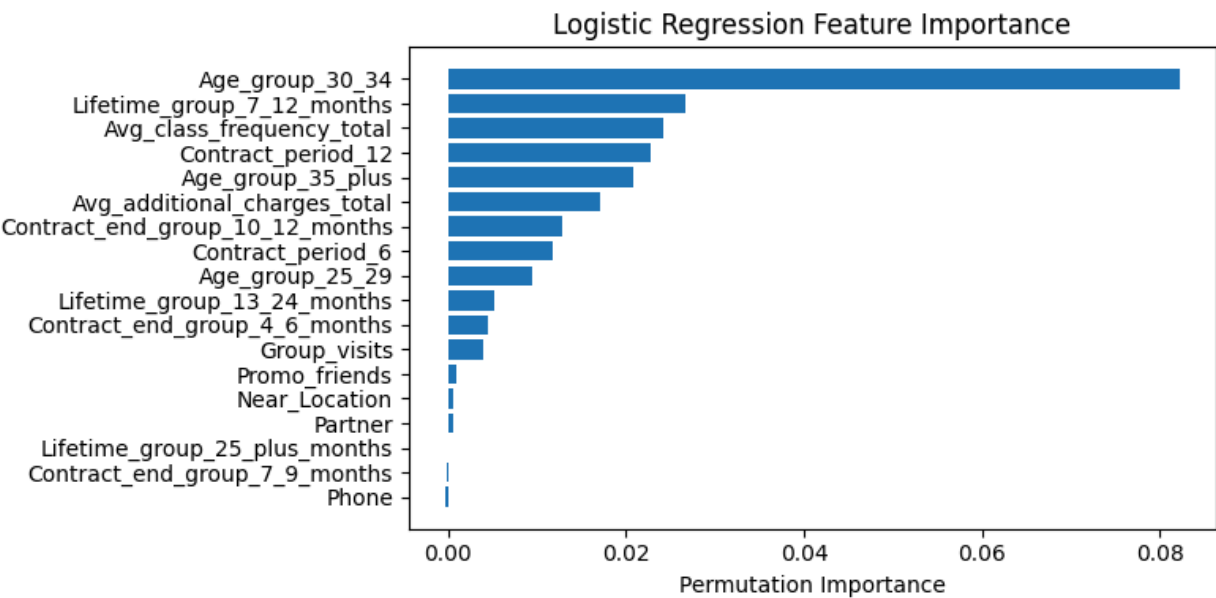
Figure 3: Model Evaluation Table

	Model	Accuracy	ROC AUC	PR AUC
0	Logistic Regression	0.840833	0.899334	0.744831
1	Decision Tree	0.825833	0.851848	0.592577
2	Random Forest	0.840833	0.883359	0.691615
3	XGBoost	0.815833	0.885795	0.692903

Model Results

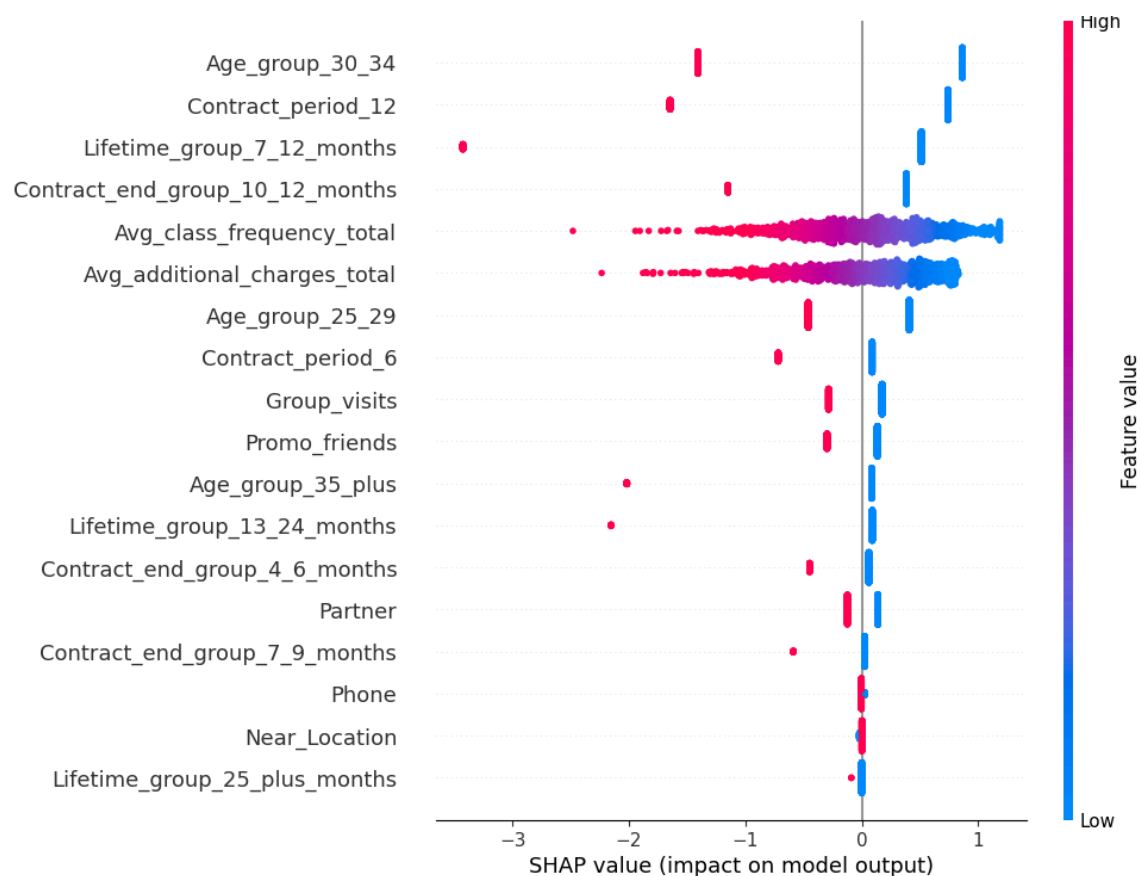
We compared Logistic Regression, Decision Tree, Random Forest, and XGBoost using accuracy, ROC AUC, and PR AUC. Logistic Regression performed best across metrics, with the strongest ROC AUC and PR AUC, while Random Forest and XGBoost trailed slightly. The Decision Tree performed the worst, especially on PR AUC, showing limited ability to identify churn. Overall, the results suggest that the dataset is best captured by a linear decision boundary, making Logistic Regression the strongest model.

Figure 4: Logistic Regression Importance Bar Chart



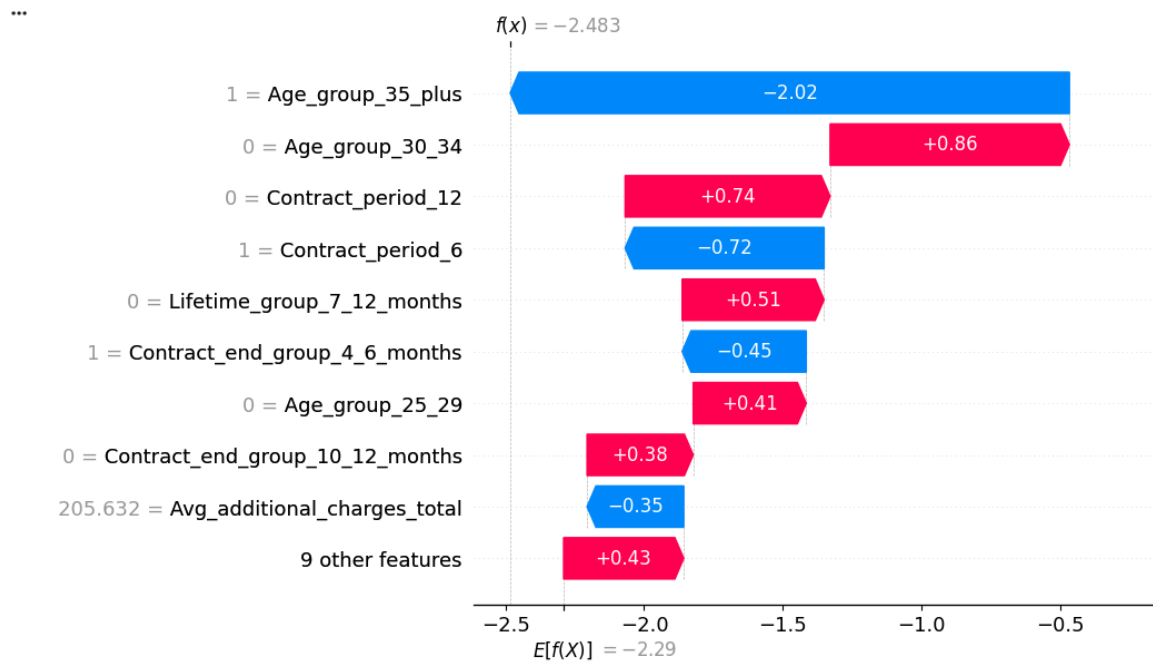
We used permutation importance to create a bar chart that gave us the analysis on which predictors played the biggest role in determining churn data. The 3 predictors that the model determined were the most important were Age group 30-34, Lifetime group 7-12 months, and Average class frequency total. Age group 30-34 was the most important predictor, indicating that people in this age range have a higher likelihood of churn, identifying that age range as a key demographic factor for retention strategies. The lifetime group 7-12 months was the second most important predictor. This predictor shows that members who have been at the gym for 7–12 months are important in determining churn, meaning that their commitment to retaining their membership is uncertain. Finally, the third most important predictor is the Average class frequency total. This means that members who are less likely to attend class are much more likely to cancel their membership.

Figure 5: Logistic Regression Beeswarm SHAP Plot



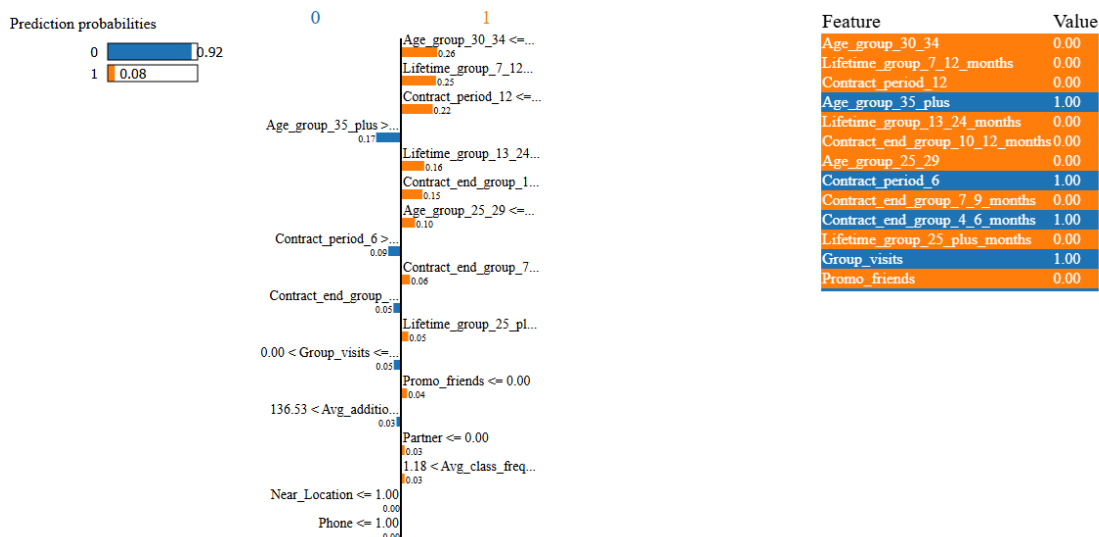
Higher feature values, such as older age groups or longer contract periods, push predictions toward non-churn, while lower feature values, such as low class frequency or shorter remaining contract time, push predictions toward higher churn risk. Overall, the plot highlights which features have the strongest impact on churn and whether high or low values of those features drive customers toward canceling or remaining.

Figure 6: Logistic Regression Waterfall SHAP Plot:



Being in the 35+ age group and having a 6-month contract reduces the predicted likelihood of churn, while features such as being in the 30–34 age group benchmark and higher additional charges increase churn risk.

Logistical Regression LIME Explainer Chart:



The model predicts only an 8% chance of churn because of certain factors. The Age group 35 plus, having a Contract period of 6, and belonging to the Contract end group 4-6 months, pushes the prediction toward non-churn. Although features like Age group 30-34, Lifetime group 7-12 months, and Contract period 12

slightly increase churn risk, resulting in an overall 92% probability that this customer will remain a member.

Discussions:

Our analysis revealed clear behavioral, demographic, and contractual patterns that explain why members choose to cancel their gym memberships. Engagement metrics carried the strongest influence. Members with low class attendance or limited overall usage were consistently more likely to churn. This highlights the importance of maintaining regular customer involvement, as disengagement often signals declining commitment. Contract structure also played a major role. Short term or monthly plans, along with contract periods nearing expiration, were closely tied to higher churn. These members appear more flexible and less committed, making them more sensitive to changes in motivation, schedule, or perceived value. Longer contract periods and members in later contract stages showed notably better retention outcomes. Demographic factors showed meaningful differences as well. Members in the thirty to thirty four age range emerged as a higher risk group, suggesting this group may face lifestyle or financial shifts that impact membership commitment. In contrast, older members, particularly those thirty five and above, were among the most stable segments, often showing consistent engagement and lower churn probability.

These insights point to several actionable strategies. First, high risk groups can benefit from personalized engagement efforts. This may include tailored class reminders, rewards for consistent attendance, or periodic wellness check ins for members who have reduced their usage. Second, improving contract retention can be supported through structured renewal incentives, discounts for longer commitments, or value added programs for members approaching contract expiration. Finally, demographic targeting can help allocate resources toward groups most likely to leave, ensuring marketing and communication efforts reach members who need support.

Overall, the combination of behavioral patterns, contract characteristics, and demographic insights gives your gym a strong foundation for developing targeted and effective retention strategies. Continued monitoring of usage trends and membership patterns will help your team make informed decisions that reduce churn and support long term customer relationships.